Business Understanding: Cross-Selling Strategy (United India Insurance)

Cross-selling means offering additional products or services to existing customers to increase revenue or strengthen customer relationships.

Objectives

1. Increase income from existing clients.
2. Protect client relationships by offering more value.

Key Points

- Cross-selling has some risk—it can affect existing relationships if not done carefully.
- The new offer should add value for the customer.
- It's often combined with up-selling (offering higher-value products).
- Benefits include higher revenue, better efficiency, and stronger customer loyalty.

Cross-Selling Cases

1. **New Coverage is greater than the Current Coverage(NC > CC)**
   → Customer gets additional or higher insurance coverage.
   (Example: ₹10L current + ₹30L new = ₹40L total)
2. **New Coverage = Current Coverage**
   → Old policy expired; offer a new one with same coverage.
   (Example: ₹10L expired → new ₹10L policy)
3. **New Coverage is less than Current Coverage(NC < CC)**
   → Old policy expired; offer new policy with smaller coverage.
   (Example: ₹40L old → new ₹20L policy)

Types of Cross-Selling

a) Identify new customer needs and offer matching products.
b) Add-on services to enhance existing products.
c) Selling solutions (e.g., product + service package).

Benefits

- Builds customer loyalty
- Strengthens relationships
- Increases sales revenue
- Improves customer satisfaction
- Raises Customer Lifetime Value (CLV)

Customer Lifetime Value (CLV)

Represents the total money a customer is expected to spend during their relationship with the company.

CLV = (Average Purchase Value) × (Average Purchase Frequency)

Insurance Services Offered

- Motor
- Health
- Travel
- Personal Accident
- Householder's
- Shopkeeper's

Data Understanding/Description

- **Age:** Represents customers' age, ranging from 18 to 60 years.
- **Gender:** gender of the customer — *Male* or *Female*.
- **Marital Status:** whether a customer is Single, Married, or Divorced.

- **Family Members:** Represents the number of family members, ranging from 1 to 10.
- **Education:** Shows the educational background of the customer.
- **Occupation and Job Title:** Provides details about the profession and job designation of each customer.
- **Income**: Lists the annual income (in lakhs of rupees) of customers.
- **Current Product**: Indicates whether the customer currently holds an insurance policy. This helps distinguish between new and existing customers.
- **Current Product Type**: Represents the product code or type of the customer's current insurance policy.
- **Current Coverage**: Specifies the coverage amount (in rupees) under the current insurance policy.
- **New Product Type**: Represents the product code or type of the new insurance product offered to the customer as part of the cross-selling campaign.
- **New Coverage**: Shows the total insurance coverage (in rupees) a customer holds after opting for a new product. For existing customers, this is the cumulative sum of the current and new product coverages.
- **Rating**: Reflects the customer's level of engagement or satisfaction with the company's services, categorized as Cold, Warm, or Hot.
- **Converted**: Indicates whether the cross-selling effort was successful — i.e., whether the customer purchased the new insurance product.
- **Status:** Similar to "Converted", but provides specific company-assigned codes explaining non-conversion reasons based on internal policy.

NUMERICAL COLUMNS / CATEGORICAL --UNIVARIATE

AGE:(Histrogram/boxplot)

-missing ages are 44.

- customers ages are in range between(18-60)years.

-mean age of customers is 39 years.

-younger customers below 25 have less insurance awareness, maximum ages are lies in between 35 to 50 years are the targeted customers.

FAMILY MEMBERS: (Histrogram/countplot)

-missing value 22

-in family 2-5 members prefer the insurance plan ,more than 6 members they the highest targeted customers

-there is no outliers.

INCOME : (Histrogram)

- Income from 3 – 9 lakhs are more interested customers.
- It has positive distribution (right skewed).
- High income people focus on the investment as compared to low income

CURRENT COVERAGE : (histrogram/boxplot)

- Missing data 49
- Heavy skewed
- It has outliersMean =3.633118 and median =50000,median is used to impute the missing value

NEW COVERAGE: (histrogram)

-missing value -127

- ranging from 10-15 lakhs

- no outliers

GENDER: (countplot)

- Missing value – 182
- Females = 39.894548 and males = 60.105452
- Value count of male is more

EDUCATION: (countplot)

BD-29586 LHS-13399 MD- 22543 NE-7232 PD-14746 UHS-12448

- Only BD and PD are target
- Missing values are imputed by mode

OCCUPATION: (countplot)

-the occupation are SPT,SE,SFT

- only SPT=52.0294 and SE=39.4453 are the targeted customers

- missing values are imputed by mode

JOB TITLE: (countplot)

- Missing data- 123
- BA-7980, CB-5396, CF-2142, DD-9795 ,FH-23780 ,OC-1660, OM-5935 , OT-6530 ,PA-1609, PG-19623 ,PR-7666, RR-7761
- Most interested are FH ,PG

RATING:

- missing values- 22
- Cold-47551, Hot-22988 ,Warm-29417
- Missing Values were imputed with the mode
- Hot customers are highly converters, warm customers is average,cold are not interested.

CURRENT PRODUCT TYPE:

- Missing data- 33
- ANS-18461 END-5487 INV-10346 NA-42855 PMT-8477 TLE-14341
- Only the ANS are target customers
- Missing value calculated by mode

NEW PRODUCT TYPE:

- Missing data- 47
- ANS-29151 END-12718 INV-20429 PMT-20429 TLE-25067

CONVERTED:

- Missing data- 49
- More are not converted
- Converted= 38296 ,not converted= 61655

STATUS:

- No missing data
- It shows the less are converted

MARITAL STATUS:

- Missing data- 42
- Divorced-23160 , Married-53950 , Single-22848
- More are married
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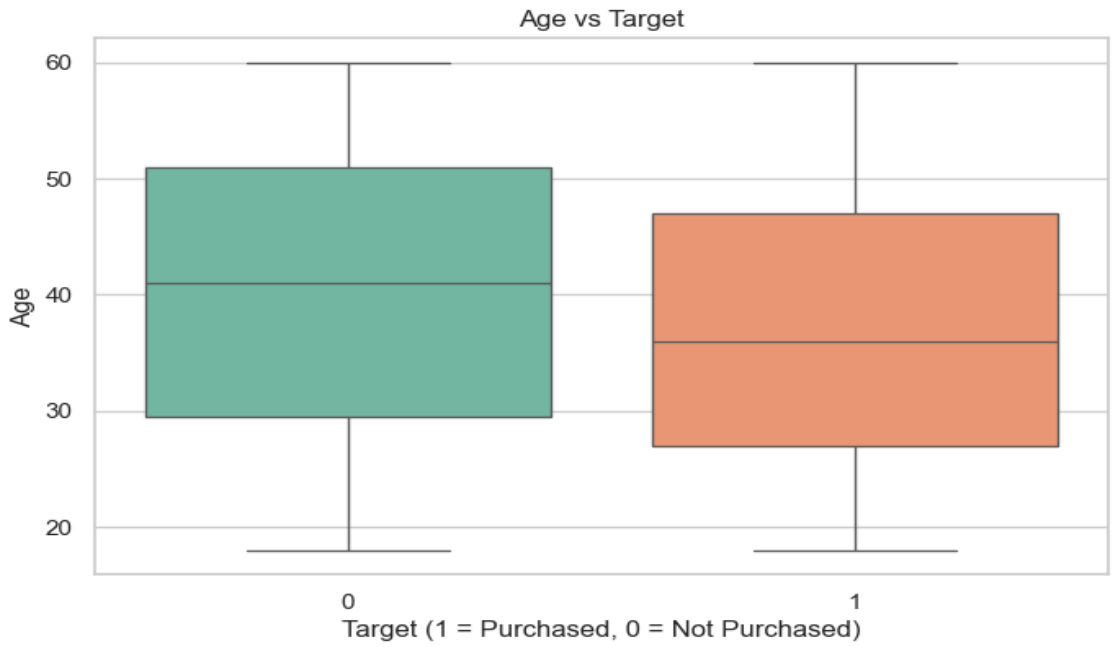
BIVARIATE ANALYSIS

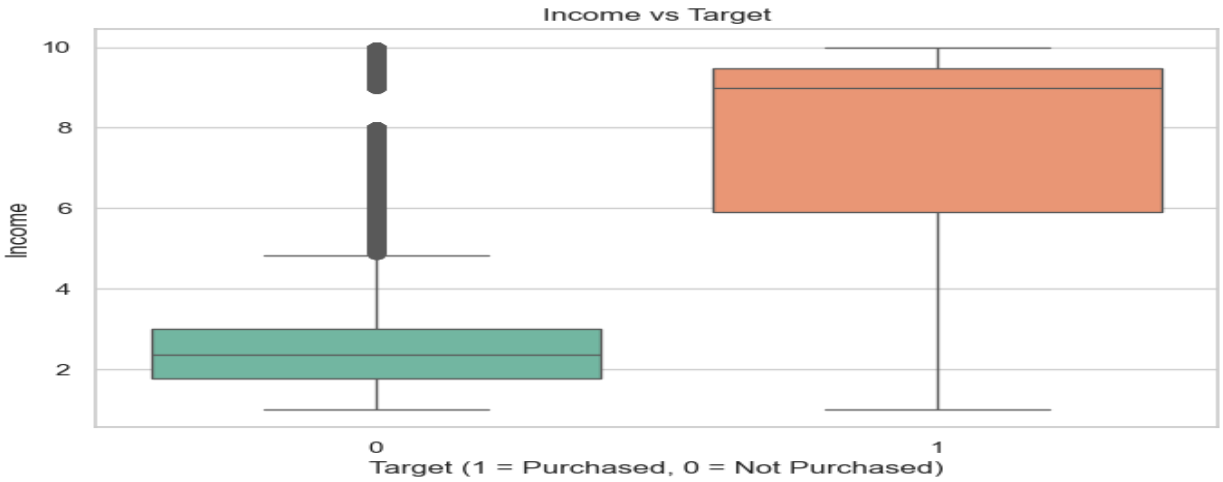
NUMERICAL COLUMNS VS TARGET

AGE VS TARGET



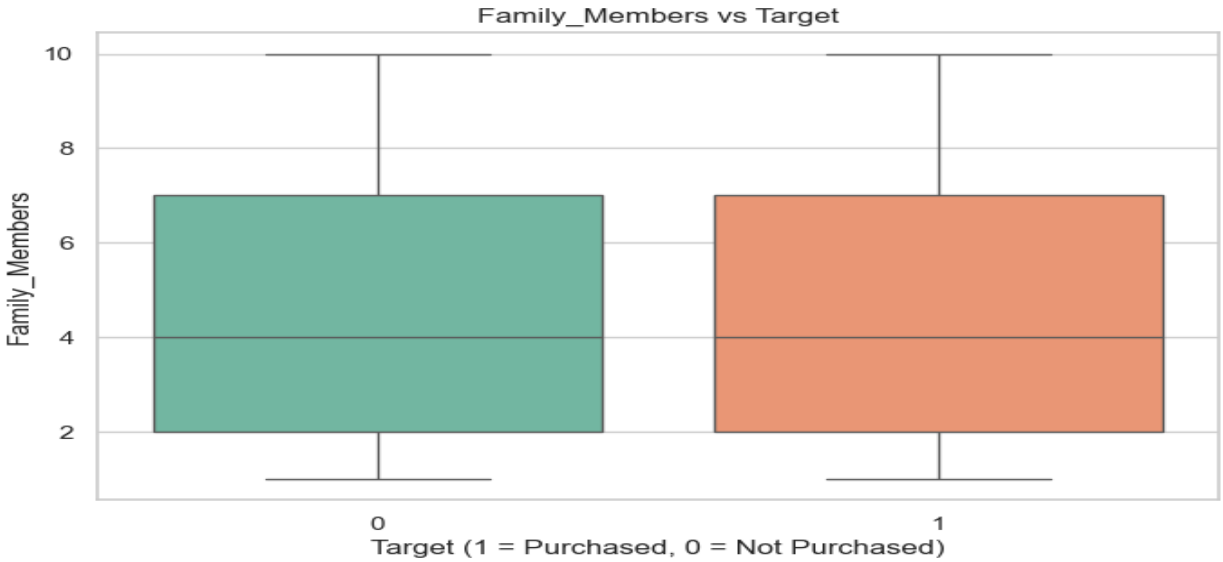
- The minimum age is 18 years and maximum age 60 years.
- The average mean age is 39 years, shows the mid age group are likely to invest in insurance
- The mid age customers knows how to manage their financial responsibility .

INCOME VS TARGET



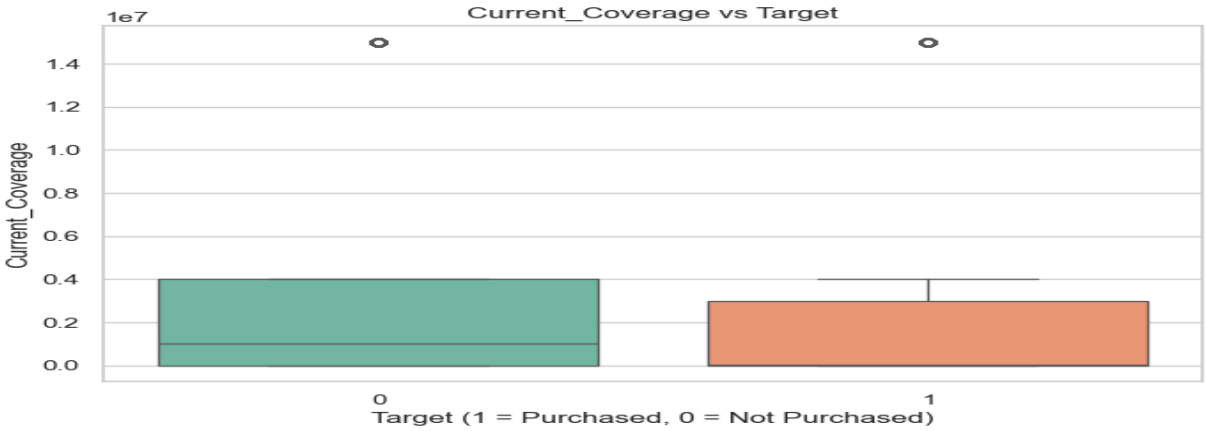
- Customers who have more income they can investment in insurance
- Low income customers are only focus on less term
- It has so many outliers who are not purchased the product

FAMILY MEMBERS vs TARGET



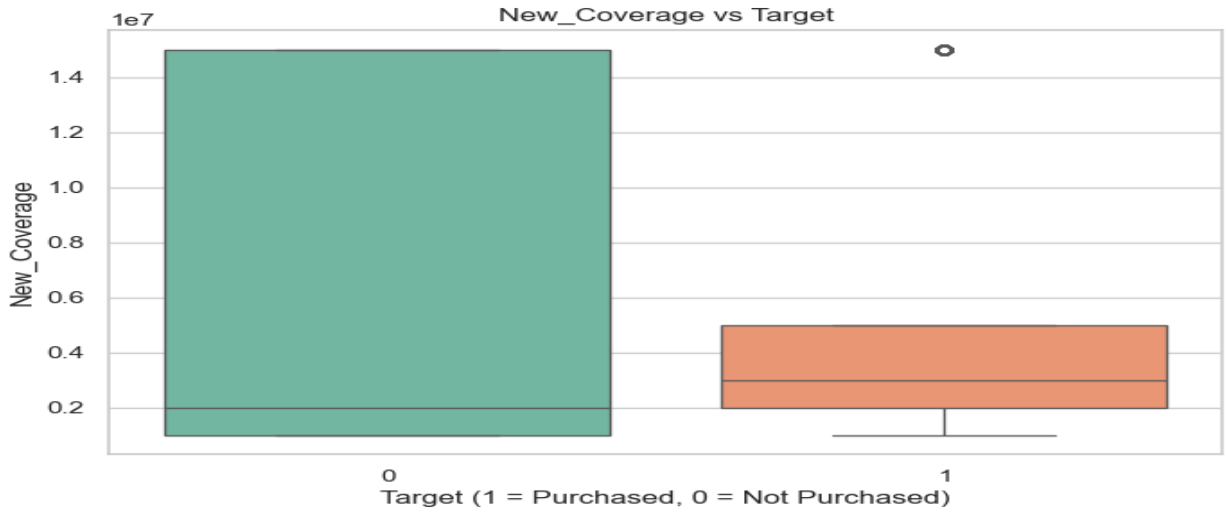
- Targeted customers are upto 2 -4 family members
- Out of 1 lakhs customers converted and not converted are equal
- Converted=4.65 ,not converted=4.66

CURRENT COVERAGE VS TARGET



- In existing customers some have the coverage
- There are outliers
- Existing customers can be taken up for cross selling
- Converted =2730547 , not converted =41933735

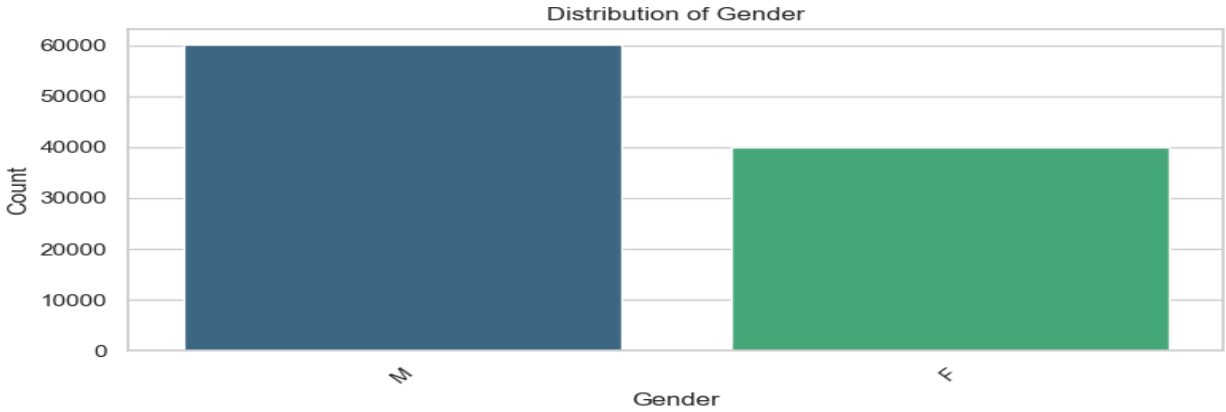
NEW COVERAGE VS TARGET



- In this more are not converted
- If they force to this new customer for cross selling they might lose their customer trust.
- No of converted is minimum =55957895
- No. of not converted is maximum= 6421539

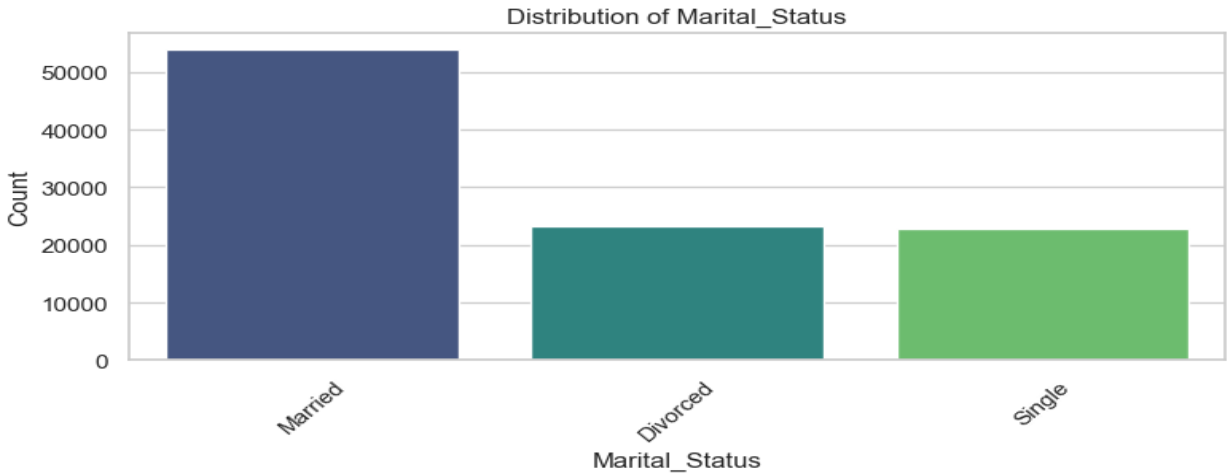
CATEGORICAL COLUMNS VS TARGET

1. AVERAGE CONVERSION RATE BY GENDER



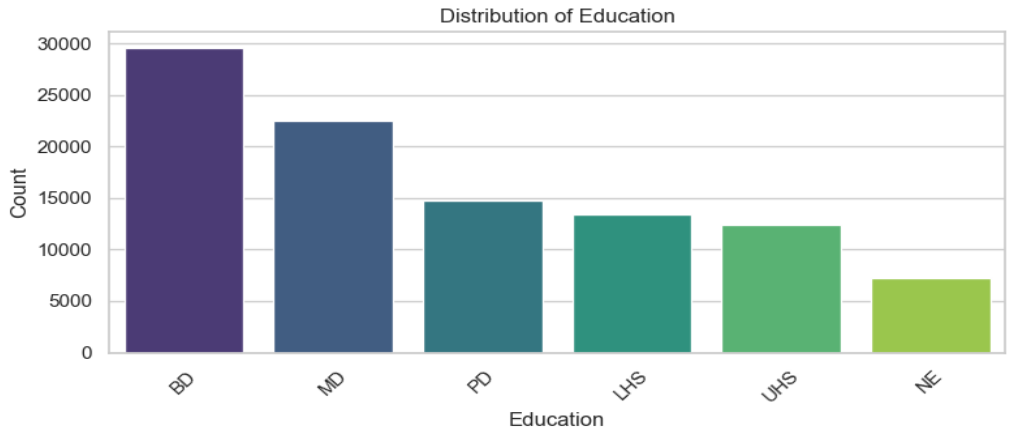
- Out of 1 lakhs customers – 38.32 are the female and males are 38.31 are the average conversion
- Males are more because they are one who mostly take care of the families and have the responsibilities .
- Females are also converting if they are the only one who take responsibilities for family

2- MARITAL STATUS



- In marital status they are 3 category -married,divorced,single
- Married= 37.81, divorced= 35.98, single= 41.89
- Most converted one is married customers because they buy the policies for family protection, single and divorced due to less financial responsibilities.

3 -**EDUCATION**

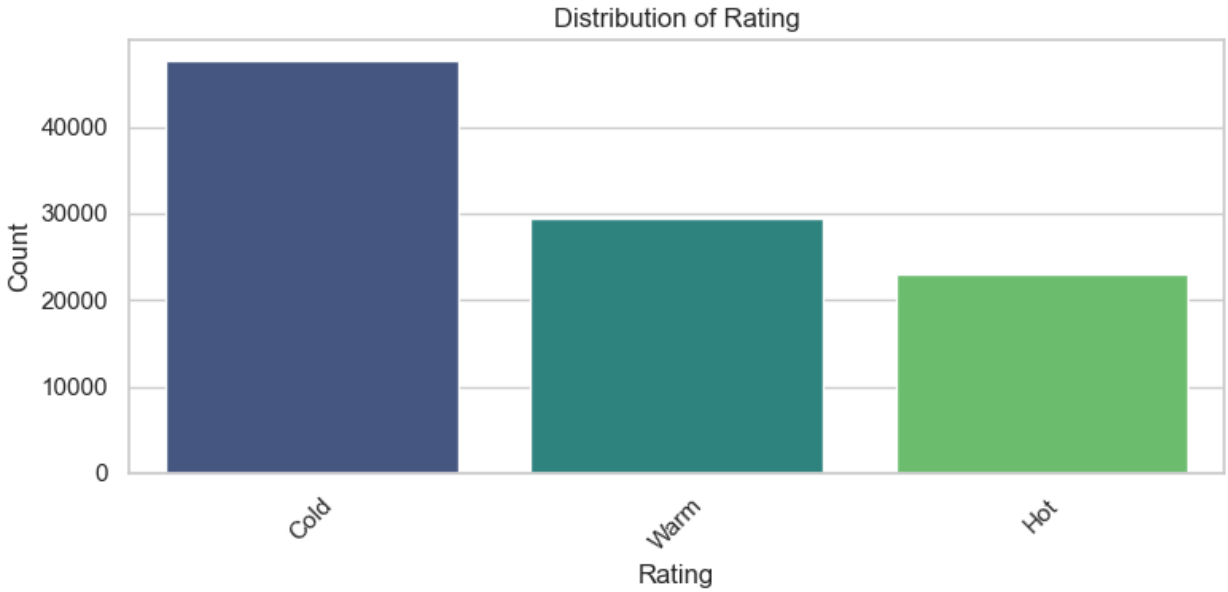


- The educated people are aware of the insurance and they willing to understand the policies.
- Otherhand ,the one who don't have proper education they cannot understand the policy properly and always have concerned about their money .
- PD=49.44,MD=49.00 ,BD=48.77 these 3 are the converted .
- Rest NE=23.29 ,LHS=15.44,UHS= 14.25 are not converted .

OCCUPATION

- Only 3 occupation are SPT=60.91,SFT=19.96,SE=12.48.
- SPT has the higher conversion rate due to better income
- Low conversions can be targeted with offers to invest in the policies.

RATING. -



- The ratings are hot=89.87,warm=52.15,cold=4.87

- Hot-rated customers (high satisfaction) are more likely to convert**.**
- Cold-rated customers show low conversions**,** meaning they may need better service or offers.
- The most focused on hot and warm rating and target the customers for cross selling.